

to use it for applying subtler effects. This actuator would be suitable when the device only needs to alert a user urgently when a certain event occurs. ERM motor also suffers from lower response times as it takes a little extra time for the vibration to begin or end.

Linear resonant actuators

The latest smartphones use linear resonant actuators (LRAs) that are capable of providing a more precise and softer vibration that does not feel jarring in your pants. In an LRA, you will find a magnet attached to a spring which is surrounded by a coil. As the magnet is controlled, it oscillates linearly until the frequency reaches the resonant frequency of that component. Frequencies on either side of the resonant frequencies can also be used at the loss of performance. On the brighter side, the LRA is more power-efficient than the ERM motor. However, it also has a

Components of haptic devices

PicoVibe 304-002. A small eccentric rotating mass motor from Precision Microdrives based on neodymium (an exotic metal) magnets for improved performance. Its size is mere 4 mm.

Fairchild Semiconductor's FAH4830. A haptic driver that supports both ERM motors and LRA and features very low wake-up time of 30 μ s.

Texas Instruments' DRV8601. A haptic driver that supports both ERM motors and LRA. It is a low-power-consumption model that consumes just 10 nA of shutdown current.

Texas Instruments' DRV8662. A haptic driver that supports piezo actuators and has an integrated voltage boost converter to account for the high voltages required of piezoelectric actuators.

PrecisionVibe C10-100. An LRA-based actuator with 1.4G amplitude. It is based on a brushless design that gives it an enhanced working life due to the reduction of mechanical components compared to the ERM-based motors.

weakness—linear vibrators are much larger than rotational vibrators.

Piezoelectric actuators

Piezoelectric actuators, or piezos as they are lovingly called, are yet another technology that allows very power-efficient vibrations. Most of you must be familiar with the piezoelectric buzzer used in electronics projects. Well, this is almost the same thing. While piezoelectric actuators allow a far wider range of frequencies to be put on the device, these also require a far higher voltage than an ERM or LRA.

The DRV8662 is a single-chip piezo haptic driver from TI that solves the high voltage requirements of piezos by including an integrated on-chip boost converter which can boost up to 105 V.

Although piezos require far higher voltage than ERM and LRAs, their extremely quick response rate results in a more power-efficient performance than the others.

Haptic drivers

Another component that is needed for imple-

menting haptics is the haptic driver. The driver is instrumental in providing a touch sensation when the user interacts with a device. It is also capable of creating a more local effect and assures the user of a correct 'point-of-touch' on the screen.

Some of the newer drivers, like the FAH4830 haptic driver from Fairchild Semiconductor, are capable of driving both ERM and LRA motors while also performing at a very low wake-up time of less than 30 μ s. Quick wake-up time is important if the user experience holds value in the system designer's plans, as it provides low latency and thus creates a faster and more realistic haptic response.

New haptic drivers also fall in line with the general consensus of design engineers that power efficiency is vital to the success of a product. Hence the latest drivers feature very low current consumption—the FAH4830 uses less than 500 nA of current on standby. TI also has an interesting haptic driver known as the DRV8601, which works on a quiescent current of 1.7 mA and a shutdown current of 10 nA while featuring a turn-on time of 100 μ s.

Implementing a different haptic technique

Next-generation actuator technologies focus more on being able to deliver a wider range of effects by manipulating the response times and frequency of the vibration. One of the most interesting techniques for effecting haptics is

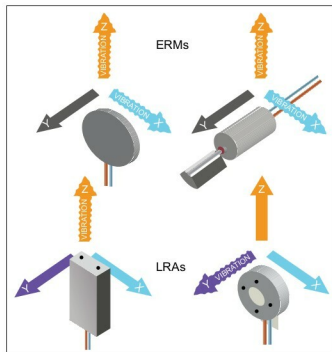


Fig. 1: ERM and LRA components

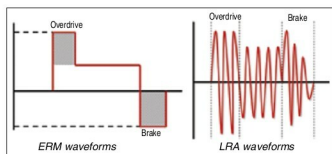


Fig. 2: Drive waveforms